



Hepatitis Viruses — Hepatitis A and E — MCQ Practice

Microbiology GIT Block · MEDucation by Mattin Majid

HOW TO USE THIS SET

20 questions, 5 options each (A–E). Answer on paper or in your head, then check every answer against the key on the final page.

4 Easy · 10 Medium · 6 Hard · suggested time ≈ 25 minutes.

Q Questions**1. What does the word "hepatitis" literally mean?**

EASY

- A. Infection of the kidney
- B. Inflammation of the pancreas
- C. Enlargement of the spleen
- D. Infection of the gallbladder
- E. Inflammation of the liver

2. Which family does Hepatitis A virus belong to?

EASY

- A. Picornaviridae
- B. Hepadnaviridae
- C. Flaviviridae
- D. Hepeviridae
- E. Caliciviridae

3. What is the structural composition of Hepatitis A virus?

EASY

- A. Naked (non-enveloped)
- B. Enveloped with a lipid bilayer
- C. Enveloped with a glycoprotein coat only
- D. Partially enveloped
- E. Enveloped with a double membrane

4. What is the route of transmission for Hepatitis A virus?

EASY

- A. Enteric (fecal-oral)
- B. Parenteral (blood-borne)
- C. Airborne droplet
- D. Vector-borne (mosquito)
- E. Sexual contact only

5. Which hepatitis viruses are transmitted via the parenteral route?

MEDIUM

- A. Hepatitis A and E only
- B. Hepatitis B, C, and D
- C. Hepatitis A, B, and E
- D. All hepatitis viruses equally
- E. Hepatitis A and D only

6. Which hepatitis viruses can progress to chronic infection?

MEDIUM

- A. Hepatitis A and E
- B. All hepatitis viruses equally
- C. None of the hepatitis viruses
- D. Hepatitis B, C, and D
- E. Hepatitis A only

7. Which hepatitis viruses are considered oncogenic?

MEDIUM

- A. Hepatitis A and E
- B. Hepatitis D and E
- C. All hepatitis viruses equally
- D. Hepatitis B and C
- E. Hepatitis A only

8. Approximately what percentage of adults infected with HAV develop icteric (jaundiced) disease, compared to children?

MEDIUM

- A. 5–20% of adults, versus 75–90% of children
- B. 75–90% of adults, versus 5–20% of children
- C. Equal rates in adults and children (~50%)
- D. Icteric disease never occurs in either group
- E. Less than 5% in both groups

9. What proportion of HAV infections in children are inapparent (subclinical)?

MEDIUM

- A. 10–25%
- B. 80–95%
- C. 50–60%
- D. Less than 5%
- E. 100%, HAV is always asymptomatic in children

10. What is the approximate mortality rate of HAV infection in adults versus children?

MEDIUM

- A. 0.1% in adults versus 0.3–2.1% in children
- B. Equal mortality of about 10% in both groups
- C. Mortality approaches 50% in both groups
- D. No mortality is ever reported for HAV
- E. 0.3–2.1% in adults versus 0.1% in children

11. Which antibody class indicates acute/current Hepatitis A infection on serologic testing?

MEDIUM

- A. Anti-HAV IgM
- B. Anti-HAV IgG
- C. Anti-HBs
- D. HBsAg
- E. Anti-HBc IgG

12. Which two forms of immunization are mentioned for Hepatitis A prevention?

MEDIUM

- A. Only active immunization is available
- B. Only passive immunization is available
- C. Passive immunization (immunoglobulin) and active immunization (vaccine)
- D. Neither passive nor active immunization exists for HAV
- E. Oral antiviral prophylaxis only

13. In which regions is Hepatitis E described as endemic?

MEDIUM

- A. Sub-Saharan Africa and South America exclusively
- B. Northern Europe and Scandinavia
- C. East Asia exclusively
- D. India and the Middle East
- E. North America exclusively

14. What dietary exposure is specifically mentioned as a prevention measure for Hepatitis E?

MEDIUM

- A. Avoiding unpasteurized dairy
- B. Avoiding undercooked pork
- C. Avoiding contaminated seafood
- D. Avoiding raw eggs
- E. Avoiding unwashed citrus fruit

15. Which statement correctly contrasts HEV with HAV regarding chronic carriage?

HARD

- A. Unlike HAV, HEV commonly causes chronic carriage
- B. HEV chronic carriage occurs only in children
- C. Like HAV, HEV chronic carrier state does not occur
- D. HEV chronic carriage occurs only in the elderly
- E. HEV chronic carriage occurs only in pregnancy

16. A pregnant woman develops HEV infection. What distinguishes her risk from a non-pregnant adult with the same infection?

HARD

- A. Her risk is identical to that of any other adult
- B. She has an increased risk of acute liver failure with high mortality, a risk specific to HEV in pregnancy
- C. She is actually at lower risk than non-pregnant adults
- D. This increased risk is specific to HAV, not HEV
- E. She has an increased risk of chronic carriage, not acute liver failure

17. Why is the naked (non-enveloped) structure of HAV and HEV clinically significant for their transmission route?

HARD

- A. It makes them more susceptible to detergents, favoring airborne spread
- B. It has no bearing on transmission route
- C. It makes them require a lipid envelope for infectivity, favoring blood-borne spread
- D. It confers resistance to heat, acid, and environmental degradation, suiting feco-oral spread through contaminated food/water
- E. It increases their susceptibility to stomach acid, preventing feco-oral spread

18. A patient has a positive anti-HAV IgG and negative IgM. What does this most likely indicate?

HARD

- A. Current acute HAV infection
- B. Active HAV viral shedding requiring isolation
- C. Vaccine failure requiring revaccination
- D. Chronic HAV carrier state
- E. Past HAV infection or immunity, not current acute infection

19. Why does this lecture's comparison table list Hepatitis D as "unknown" for oncogenicity while listing HBV and HCV as clearly oncogenic?

HARD

- A. HDV has definitively been proven non-oncogenic
- B. HDV is never found in humans, so oncogenicity cannot be assessed
- C. HDV is a defective virus dependent on HBV coinfection, and its independent oncogenic potential is less clearly established in the material presented
- D. HDV causes cancer in 100% of cases, making "unknown" an error
- E. HDV is identical to HBV, so its oncogenicity is automatically the same

20. Given that this lecture's detailed content covers only HAV and HEV, what should a student specifically avoid assuming about HBV, HCV, and HDV from this material alone?

HARD

- A. That HBV, HCV, and HDV exist at all
- B. That HBV and HCV can cause chronic infection
- C. That parenteral transmission applies to HBV, HCV, and HDV
- D. That HBV and HCV are listed in the comparison tables at all
- E. That the comparison-table facts (genome, transmission, chronicity, oncogenicity) represent the full clinical picture — detailed pathogenesis, diagnosis, and management for these three were not covered here

✓ Answer key

Q	Answer	Difficulty	Why
1	E	EASY	Hepatitis combines "hepat" (liver) and "itis" (inflammation).
2	A	EASY	HAV belongs to the Picornaviridae family (picornavirus group of enteroviruses).
3	A	EASY	HAV is a naked, non-enveloped virus, contributing to its environmental and acid stability.
4	A	EASY	HAV is transmitted via the feco-oral (enteric) route.
5	B	MEDIUM	HBV, HCV, and HDV are parenterally transmitted, in contrast to the enteric HAV and HEV.
6	D	MEDIUM	HBV, HCV, and HDV are "often" chronic, while HAV and HEV never become chronic.
7	D	MEDIUM	HBV and HCV are oncogenic (linked to hepatocellular carcinoma); HDV oncogenicity is listed as unknown.
8	B	MEDIUM	Icteric disease occurs in about 75-90% of infected adults but only 5-20% of infected children.
9	B	MEDIUM	Children have an 80-95% rate of inapparent/subclinical HAV infection, much higher than adults.
10	E	MEDIUM	Mortality is higher in adults (0.3-2.1%) compared to children (0.1%) with HAV infection.
11	A	MEDIUM	IgM anti-HAV indicates acute, current infection, while IgG indicates past infection or immunity.
12	C	MEDIUM	Both passive immunization (immunoglobulin) and active immunization (vaccine) are available for HAV prevention.
13	D	MEDIUM	HEV is specifically noted as endemic in India and the Middle East.
14	C	MEDIUM	Prevention of HEV includes better personal hygiene and avoiding contaminated seafood.
15	C	HARD	Both HAV and HEV never progress to a chronic carrier state, unlike HBV, HCV, and HDV.
16	B	HARD	HEV in pregnancy carries a distinctly higher risk of acute liver failure and mortality, a feature that differentiates it from HAV.
17	D	HARD	Naked viruses like HAV and HEV are more resistant to heat, acid, and bile, which helps them survive gastrointestinal transit and environmental contamination routes.
18	E	HARD	IgG without IgM reflects past infection or immunity rather than acute, current infection, which is marked by IgM.
19	C	HARD	The lecture explicitly marks HDV oncogenicity as unknown/uncertain, distinct from the clearly established oncogenic potential of HBV and HCV — this is presented as stated, without further elaboration in the source material.
20	E	HARD	This lecture only develops detailed clinical content for HAV and HEV; HBV, HCV, and HDV appear solely in summary comparison tables, so students should seek a dedicated source for their full pathogenesis, diagnosis, and management.